

STAT 230 — Popcorn Experiment Written Report

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Introduction

This experiment tested two factors, oil type and kernel timing, in order to see their effects on the time it took to pop popcorn. The oils we tested were olive, vegetable, and coconut oils, and the kernel timing was a cold start, or preheated oil. Decreases in popping time are useful for efficiency in the kitchen or commercially. We hoped this experiment would answer the following three questions. Does the type of oil affect how fast the popcorn finishes popping? Does pre-heating the oil affect how fast the popcorn finishes popping? And, does the impact of pre-heating the oil change depending on the oil used?

Design & Data Collection

A. Experimental Design Description

This experiment used a balanced 2x3 factorial design. We had 5 replications per treatment combination which made for 30 total observations measuring the response variable, total popping time, in seconds. Below is the statistical model we used and the definition of each part.

$$Y_{ijk} = \mu + A_i + B_j + AB_{ij} + E_{ijk}$$

Y_{ijk} : The total popping time for replicate k under oil type i and start method j

μ : The overall mean total popping time

A_i : The effect of oil type i

B_j : The effect of start method j

AB_{ij} : The interaction effect between oil type i and start method j

E_{ijk} : The residual error

B. Power Analysis

For our experiment, we aimed to achieve at least 80% power at the 0.05 significance level with six treatment groups (2 kernel-timing levels $\times$ 3 oil types). Based on reasonable assumptions about variability, we determined that five replications per treatment would likely provide sufficient power. After running the full experiment with five replications, we estimated the within-group variance using the residual mean square from the ANOVA (MSE = 293.91). The between-group variance (250.20) was obtained by computing the variance of the six treatment-combination means. These values quantify the variation within and between groups needed for the ANOVA power calculation. Using these variance estimates, our calculated power exceeded 90%, indicating a very low risk of committing a Type II error.

```
groups = 6
n = 5
between.var = 250.1991
within.var = 293.9069
sig.level = 0.05
power = 0.9092419
```

C. Procedure Description

Before collecting data, we randomized our treatments by listing every treatment-replication combination and assigning each a unique random number from 1 to 30. We then ordered the treatments according to these random numbers, creating a fully randomized treatment sequence. This procedure helped eliminate bias or lurking variables associated with treatment order and supported the independence of our trials. In collecting our data, we followed the following procedures for each trial:

1. Place medium pot on the stove with 2T. of selected oil placed inside
2. Start the timer on our phone and turn on the heat to medium high
3. If doing the cold-start, add $\frac{1}{4}$ c. of popcorn kernels into the pot of oil; else, add 4 kernels in the pot of oil and add the rest of the kernels when the first of those 4 kernels pop; place the lid on top
4. When the kernels start to pop 3 seconds apart, shake the pot occasionally on the stove using hot pads to keep the lid on
5. When the pops are 4 seconds apart and slowing down, take the pot off the heat, stop the timer, and dump the popped kernels onto the counter top
6. Wash out the pan and wait 8 minutes for the pan and stove to cool down before doing the next batch alternating between the two identical burners



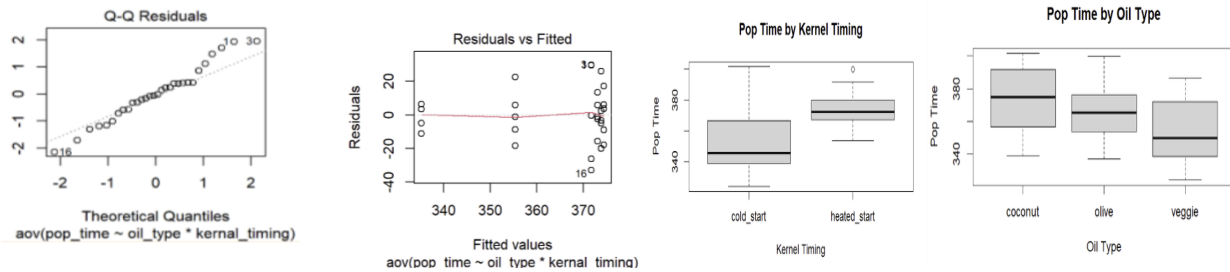
Data Analysis

A. ANOVA Table

Source	DF	Sum Sq	Mean Sq	F-value	p-value
Kernel Timing	1	2903.6	2903.6	9.879	0.00441
Oil Type	2	1798.5	899.3	3.06	0.06552
Interaction	2	1552.9	776.4	2.642	0.09184
Residuals	24	7053.8	293.9	—	—

As seen in our ANOVA table, the Kernel Timing significantly affected the total popping time with a p-value of 0.00441 at the $\alpha = 0.05$ significance level. That means that the timing of when the kernels are added has a strong impact on the total pop time. On the other hand, the Oil Type and the Interaction did not show a statistically significant effect, with p-values of 0.06552 and 0.09184 respectively.

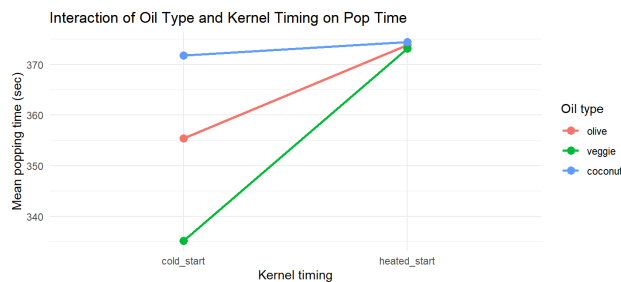
B. Assumptions checked



The Q–Q plot shows the residuals are approximately normal, and independence is supported by our randomized treatment order. The residuals-vs-fitted plot shows only a slight increase in spread; a log transform did not improve it, so we kept the untransformed model. The boxplots show similar spreads and both variance ratios are below 2, so the equal-variance assumption is reasonable.

D. Interaction Plot

At first it appeared as if there was an interaction present in the plot because of the large differences of slope in the lines that is due to the wider spread for cold start. However, our ANOVA analysis showed that this pattern was not significant. This actually confirms what the chart shows because this plot is zoomed in on only y 335 - 375 which makes the differences in slope look a lot larger than they actually are in the context of the whole. This means that preheating the oil does not have a significant impact on the impact of the different oil types.



Conclusion

In conclusion, pre-heating the oil significantly increases popping time (reject H_0 for Kernel Timing), while oil type does not have a significant effect (fail to reject H_0 for Oil). The interaction between oil type and pre-heating is also not significant, even though the interaction plot suggests a pattern. Because this was a controlled and randomized experiment, we can infer cause and effect—but only within this specific stovetop setup.

Future studies could test additional oils or heating methods, include more replications, or use equipment with controlled temperatures to reduce variability and increase power. Measuring outcomes such as unpopped kernels, popped volume, or burn rate could also give a fuller picture of popping efficiency. Using a single burner with more time between trials could further minimize variability.

Appendix

R Code:

Randomization: `sample(1:30, 30, replace = F)`

Power: `library(dplyr)`

```
Popcorn_Data$kernal_timing <- as.factor(Popcorn_Data$kernal_timing)
```

```
Popcorn_Data$oil_type <- as.factor(Popcorn_Data$oil_type)
```

```
model <- aov(pop_time ~ kernal_timing * oil_type, data = Popcorn_Data)
```

```
anova_tab <- summary(model)[[1]]
```

```
MSE <- anova_tab["Residuals", "Mean Sq"]
```

```
sigma_within <- sqrt(MSE)
```

```
cell_means <- Popcorn_Data |>
```

```
  group_by(kernal_timing, oil_type) |>
```

```
  summarise(mean_pop = mean(pop_time), .groups = "drop")
```

```
sigma_between <- sd(cell_means$mean_pop)
```

```
f_effect <- sigma_between / sigma_within
```

```
power_result <- power.anova.test(
```

```
  groups = 6,
```

```
  n = 5,
```

```
  between.var = sigma_between^2,
```

```
within.var = sigma_within^2,  
sig.level = 0.05  
)
```

Power_result

Boxplot: model <- aov(pop_time ~ kernal_timing * oil_type, data = df)

```
boxplot(pop_time ~ kernal_timing,  
        data = df,  
        main = "Pop Time by Kernel Timing",  
        xlab = "Kernel Timing",  
        ylab = "Pop Time")
```

```
boxplot(pop_time ~ oil_type,  
        data = df,  
        main = "Pop Time by Oil Type",  
        xlab = "Oil Type",  
        ylab = "Pop Time")
```

Histogram: ggplot(popcorn, aes(x = pop_time)) +
geom_histogram(binwidth = 15, fill = "skyblue", color = "black") +
labs(title = "Histogram of Kernel Timing",
 x = "Time (Seconds)",
 y = "Count")

Residuals: par(mfrow = c(1,2))

```
plot(fit, which = 1)
```

```
plot(fit, which = 2)
par(mfrow = c(1,1))
```

```
Interaction Plot: ggplot(pop,
  aes(x = kernal_timing,
    y = pop_time,
    colour = oil_type,
    group = oil_type)) +
  stat_summary(fun = mean, geom = "line", linewidth = 1) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  labs(x = "Kernel timing",
    y = "Mean popping time (sec)",
    colour = "Oil type") +
  theme_minimal()
```

ANOVA: # Convert variables to factors

```
Popcorn_Data$kernal_timing <- as.factor(Popcorn_Data$kernal_timing)
Popcorn_Data$oil_type <- as.factor(Popcorn_Data$oil_type)
# Two-Way ANOVA
model <- aov(pop_time ~ kernal_timing * oil_type, data = Popcorn_Data)
summary(model)
# Optional: Type II ANOVA (cleaner)
library(car)
Anova(model, type = 2)
```

Log Transformation: log_popcorn = popcorn %>%

```
mutate(log_pop_time = log(pop_time))
```

```
model = lm(log_pop_time ~ oil_type * kernal_timing, data = log_popcorn)
```

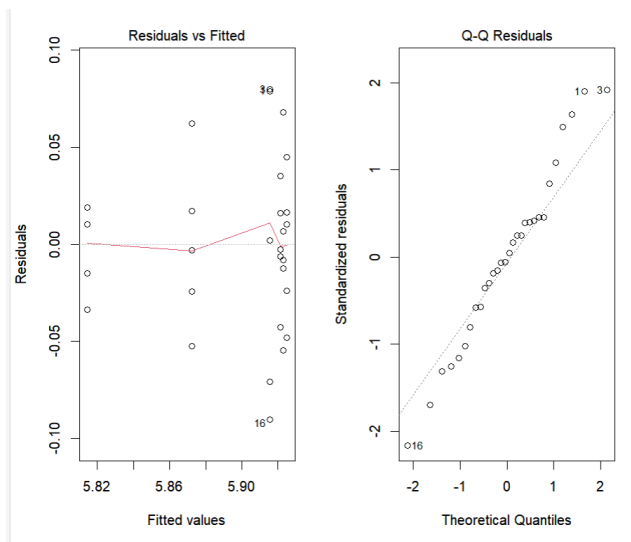
```
summary(model)
```

```
par(mfrow = c(1,2))
```

```
plot(model, which = 1)
```

```
plot(model, which = 2)
```

```
par(mfrow = c(1,1))
```

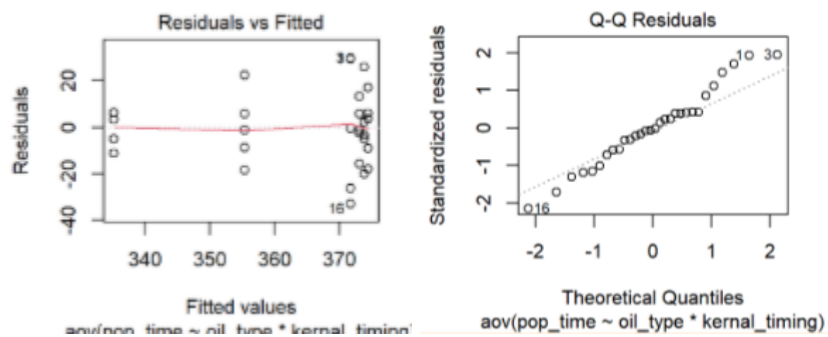


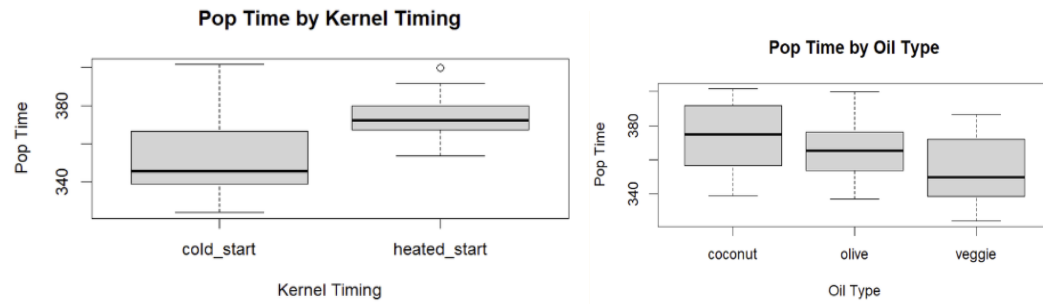
Data Table:

trial_number	kernal_timing	oil_type	pop_time
1	cold_start	coconut	401.28
2	heated_start	coconut	391.5
3	cold_start	coconut	401.59
4	heated_start	coconut	378.15
5	heated_start	coconut	380.55
6	heated_start	olive	353.71
7	heated_start	veggie	379.07
8	heated_start	veggie	357.49
9	heated_start	olive	399.83
10	heated_start	olive	368.9

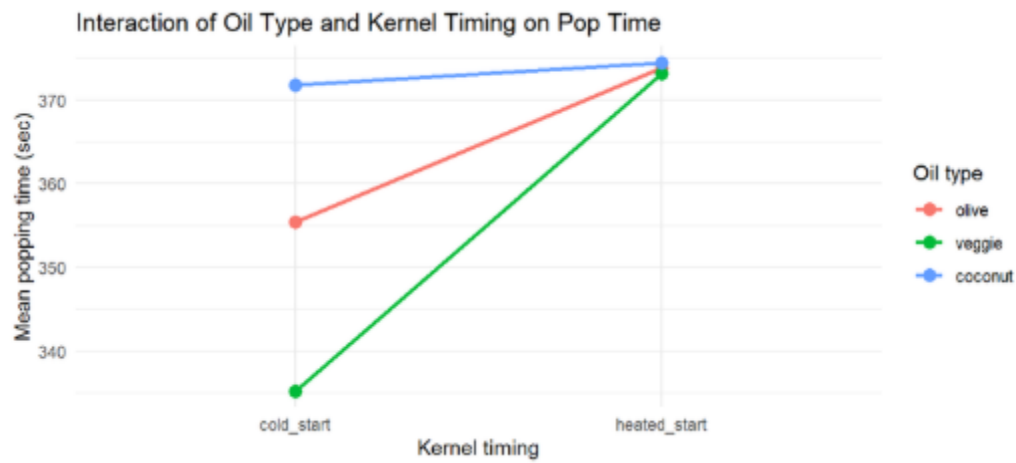
11	heated_start	veggie	370.65
12	heated_start	olive	376.06
13	cold_start	coconut	345.49
14	cold_start	olive	377.94
15	cold_start	coconut	371.5
16	cold_start	coconut	338.84
17	cold_start	olive	354.14
18	cold_start	veggie	324.06
19	cold_start	veggie	330.24
20	cold_start	olive	361.29
21	heated_start	olive	370.58
22	heated_start	veggie	372.08
23	cold_start	veggie	338.61
24	cold_start	veggie	341.6
25	cold_start	olive	346.67
26	heated_start	veggie	386.34
27	heated_start	coconut	356.68
28	cold_start	veggie	341.59
29	heated_start	coconut	365.47
30	cold_start	olive	337.08

Residual Plots:





Interaction Plots:



Photos:

